UNIVERSITY OF LIMPOPO

FACULTY OF SCIENCE AND AGRICULTURE
SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

Design Implementation and Secure Operation of a Multi Site Smart Campus Network

Final Project Report

Project detail	Information
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Assessment	Semester Project
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Declaration

We declare that this report represents the work completed for the SCOB032 Computer Networks semester project. All external sources, standards and technical documentation used in the design have been acknowledged in the reference list. Each member's contribution is recorded in the contribution statement at the end of this report.

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1 Executive Summary

This project presents the design and implementation of a secure multi-site smart campus network for the University of Limpopo using Cisco Packet Tracer. The solution connects the main campus to a satellite campus and supports academic, administrative, residential and operational services. The design separates users and services through virtual local area networks (VLANs), uses variable-length subnet masking (VLSM) to allocate address space efficiently, and provides redundant default gateways through the Hot Standby Router Protocol (HSRP). Two main-campus multilayer switches provide resilient inter-VLAN routing, while the Spanning Tree Protocol (STP) controls redundant access uplinks and prevents switching loops.

Security was implemented in layers. Extended access control lists (ACLs) enforce zero-trust-style restrictions between student, residence, research, Internet of Things (IoT), management and server networks. Port security, unused-port shutdown, Secure Shell (SSH) management and wireless IEEE 802.1X authentication reduce unauthorized access. An IPsec site-to-site virtual private network (VPN)

protects traffic exchanged between the two campuses. Central Domain Name System (DNS), Dynamic Host Configuration Protocol (DHCP), Remote Authentication Dial-In User Service (RADIUS), Network Time Protocol (NTP), syslog, web and software-defined networking (SDN) services support network operations.

The completed prototype supports wired and wireless connectivity, lightweight access points, staff and student authentication, IoT registration, Voice over Internet Protocol (VoIP) calling and priority treatment for voice traffic. Testing confirmed DHCP address allocation, VLAN reachability, ACL enforcement, HSRP failover, VPN encryption, wireless authentication, telephone registration and inter-campus calls. Cisco Packet Tracer limitations affected detailed Quality of Service (QoS) counter visibility after IPsec encryption; however, voice classification, Differentiated Services Code Point (DSCP) marking and attached priority policies were verified before and at the WAN edge.

SDN-based monitoring was demonstrated through a central controller and Python application programming interface (API) script, which reported the reachability and collection status of managed network devices. The modular VLAN structure, redundant core design and centralized services allow the network to be expanded as user demand increases. By combining simulated enterprise technologies with cost-conscious device placement, the project achieved its main objectives of security, availability, scalability and centralized management.

2 Introduction

2.1 Background

A modern university depends on reliable connectivity for teaching, research, administration, residences, security and digital services. Placing all users and devices in one broadcast domain would create avoidable congestion and expose sensitive systems to unnecessary risk. The proposed design therefore uses a hierarchical, segmented architecture that can serve more than 500 concurrent users while remaining manageable in a simulated environment.

2.2 Problem Statement

The university requires one coordinated network that connects multiple buildings at the main campus to a remote satellite campus. The network must carry data, voice, video, wireless and IoT traffic without allowing unrestricted movement between user groups. It must remain available during the failure of a gateway or uplink, protect traffic crossing the public network, provide centralized authentication and monitoring, and allow future expansion without redesigning the entire addressing plan.

2.3 Project Objectives

- Design logical and physical campus topologies using a scalable hierarchical model.
- Create VLAN and VLSM plans for users, voice, wireless, IoT, servers and management.
- Provide dynamic addressing and central services to both campuses.
- Implement resilient routing and switching using HSRP, redundant trunks and spanning tree.
- Secure the design using ACLs, zone-based firewalling, IPsec VPN, 802.1X, SSH and port hardening.
- Integrate central wireless control, SDN monitoring, IoT services, VoIP and QoS.
- Test connectivity, failover, isolation, authentication, services and performance.

2.4 Scope

The implementation covers the main campus buildings, including administration, lecture halls, the library, student residences, mathematics staff offices and research laboratories, together with the

server and ICT administration environment. The satellite campus includes staff and student networks, printers, voice services, IoT devices, staff Wi-Fi, student Wi-Fi and management networks. Inter-campus communication is represented by an Internet service provider segment, while Cisco Packet Tracer provides the routers, multilayer switches, access switches, wireless controllers, lightweight access points, servers, IP phones, IoT devices and user endpoints.

The project is limited to the logical design, configuration and testing of the network within Cisco Packet Tracer. Physical implementation activities, including structured cabling, wireless site surveys, equipment procurement and testing under real production traffic, are outside the project scope. Nevertheless, the design considers scalability, security and cost-conscious deployment, providing a practical foundation for possible future implementation in a real campus environment.

3 Requirements and Methodology

3.1 Requirements Analysis

The network requirements were derived from the operational needs of the University of Limpopo. The design was intended to support more than 500 concurrent users across two campuses while providing secure access, wireless mobility, voice and video traffic, IoT connectivity, centralized services and reliable network management. Each requirement was mapped to a technical solution for implementation and verification within Cisco Packet Tracer. These solutions included hierarchical switching, VLAN segmentation, VLSM addressing, HSRP gateway redundancy, spanning tree, an encrypted site-to-site IPsec VPN, wireless authentication, access control lists, QoS policies and SDN-based monitoring. Testing focused on representative devices and services to demonstrate connectivity, segmentation, redundancy and security. Support for 500 concurrent users remained a design objective rather than a measured simulation result.

Requirement	Implementation response
500+ concurrent users	Large student and residence subnets, VLAN segmentation and hierarchical switching
Multiple campuses	Encrypted site-to-site IPsec VPN over simulated ISP links
High availability	Dual main core switches, HSRP, redundant access uplinks and PVST
Secure access	ACLs, 802.1X/RADIUS, SSH, port security, BPDU Guard and unused-port shutdown.
Wireless mobility	WLC-managed LAPs with staff, student and residence WLANs
Voice and video	CME voice gateways, separate voice VLANs and WAN QoS policies
IoT services	Dedicated IoT VLANs and central registration server
Management	Management VLAN, NTP, syslog, SDN controller and monitoring script

3.2 Implementation Method

The team followed an iterative build-and-test method. The addressing plan and VLAN structure guided the deployment of access switches, end devices and trunk links. Layer 3 gateways, HSRP and DHCP services were then configured to provide inter-VLAN routing, gateway redundancy and automatic IP address allocation. Security controls were introduced and tested alongside basic connectivity. Wireless access, RADIUS authentication, IoT services, VoIP, the site-to-site IPsec VPN, SDN monitoring and QoS policies were integrated in stages. Connectivity tests, service checks and device command outputs were used to verify configurations and identify faults. Issues discovered during testing were investigated, corrected and retested, and the final configurations were saved for demonstration and submission.

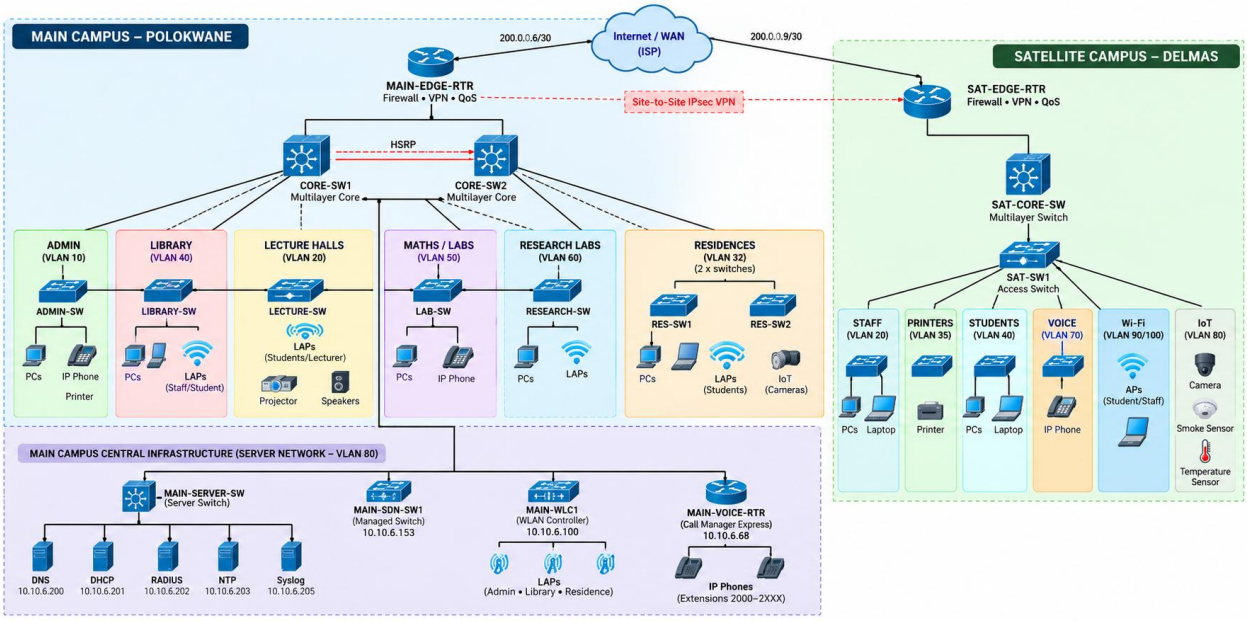


Figure 3.1: Implemented multi-site smart campus network topology in Cisco Packet Tracer.

4 Network Architecture

4.1 Main Campus

The main campus follows a collapsed-core design in which CORE-SW1 and CORE-SW2 provide Layer 3 switching, inter-VLAN routing and HSRP gateway redundancy. Building access switches are connected to the core through redundant trunk links, allowing Spanning Tree Protocol (STP) to maintain a loop-free forwarding topology while preserving alternative paths in the event of a link failure. MAIN-EDGE-RTR connects the main campus to the simulated ISP and terminates the main-campus side of the site-to-site IPsec VPN. Central network services, wireless infrastructure, voice services, monitoring and SDN components are connected through the main-campus switching infrastructure.

4.2 Satellite Campus

The satellite campus uses SAT-CORE-SW for inter-VLAN routing and policy enforcement, while SAT-SW1 provides access connectivity for staff, students, printers, wireless access points and IoT devices. SAT-EDGE-RTR provides WAN connectivity and terminates the satellite side of the site-to-site IPsec VPN. SAT-VOICE-RTR provides local Cisco CallManager Express services for satellite IP phones, while SAT-WLC1 manages the satellite wireless infrastructure. The satellite campus maintains its own local

VLANs and default gateways while using the VPN to securely access authorized resources located at the main campus.

4.3 Device Roles

Device or group	Primary role
CORE-SW1 / CORE-SW2	Main-campus Layer 3 switching, inter-VLAN routing, HSRP, ACL enforcement and DHCP relay
MAIN-EDGE-RTR	Main-campus WAN routing, zone-based firewall, IPsec VPN termination and QoS
SAT-CORE-SW	Satellite inter-VLAN routing and policy enforcement
SAT-EDGE-RTR	Satellite WAN routing, zone-based firewall, IPsec VPN termination and QoS
MAIN-VOICE-RTR / SAT-VOICE-RTR	Cisco CallManager Express, IP phone registration and inter-campus dial peers
MAIN-WLC1 / SAT-WLC1	Central wireless LAN control and WLAN-to-VLAN mapping
Access switches	Endpoint connectivity, VLAN assignment, trunking, port security, PortFast and BPDU Guard
Central servers	DNS, DHCP, RADIUS, NTP/syslog, IoT, web and SDN services

5 IP Addressing and VLAN Design

5.1 Main Campus Addressing

VLAN	Name	Subnet	Virtual gateway	Purpose
10	ADMIN	10.10.7.0/25	10.10.7.1	Administrative users
20	STAFF	10.10.4.128/25	10.10.4.129	Academic and support staff
30	RESEARCH-LABS	10.10.7.128/25	10.10.7.129	Research laboratory computers
35	PRINTERS	10.10.6.0/26	10.10.6.1	Shared printers
40	STUDENTS	10.10.4.0/25	10.10.4.1	Wired student users
50	RESIDENCE	10.10.8.0/22	10.10.8.1	Residence students and residence WLAN
60	SERVERS	10.10.6.192/26	10.10.6.193	Central services and controllers
70	VOICE	10.10.6.64/26	10.10.6.65	IP phones and call control
80	IOT	10.10.5.128/25	10.10.5.129	Cameras and environmental sensors
90	STAFF-WIFI	10.10.5.0/25	10.10.5.1	Authenticated staff wireless
100	STUDENT-WIFI	10.10.0.0/22	10.10.0.1	Authenticated student wireless
110	MANAGEMENT	10.10.6.128/26	10.10.6.129	Infrastructure management

120	ICT-ADMIN	10.10.12.0/27	10.10.12.1	Privileged ICT administration

```
Password:
CORE-SW1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Gig0/1
                                                Gig0/2
10   ADMIN                            active
20   STAFF                            active
30   RESEARCH-LABS                    active
35   PRINTERS                         active
40   LIBRARY-USERS                    active
50   RESIDENCE                        active
60   SERVERS                          active
70   VOICE                            active    Fa0/23
80   IOT-SECURITY                     active
90   STAFF-WIFI                       active
100  STUDENT-WIFI                     active
110  MANAGEMENT                       active
120  ICT-ADMIN                        active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-SW1#
*Sep 30, 13:10:07.1010: %HSRP-6-STATECHANGE: Vlan30 Grp 30 state Speak -> Standby
*Sep 30, 13:10:07.1010: %HSRP-6-STATECHANGE: Vlan30 Grp 30 state Standby -> Active
```

Figure 5.1: Main-campus VLAN configuration on CORE-SW1.

5.2 Satellite Campus Addressing

VLAN	Name	Subnet	Gateway	Purpose
20	SAT-STAFF	10.20.20.0/24	10.20.20.1	Satellite staff
35	SAT-PRINTERS	10.20.35.0/24	10.20.35.1	Satellite printers
40	SAT-STUDENTS	10.20.40.0/24	10.20.40.1	Satellite students
70	SAT-VOICE	10.20.70.0/24	10.20.70.1	Satellite voice
80	SAT-IOT	10.20.80.0/24	10.20.80.1	Satellite IoT
90	SAT-STAFF-WIFI	10.20.90.0/24	10.20.90.1	Satellite staff wireless
100	SAT-STUDENT-WIFI	10.20.100.0/24	10.20.100.1	Satellite student wireless
110	SAT-MANAGEMENT	10.20.110.0/24	10.20.110.1	Satellite management

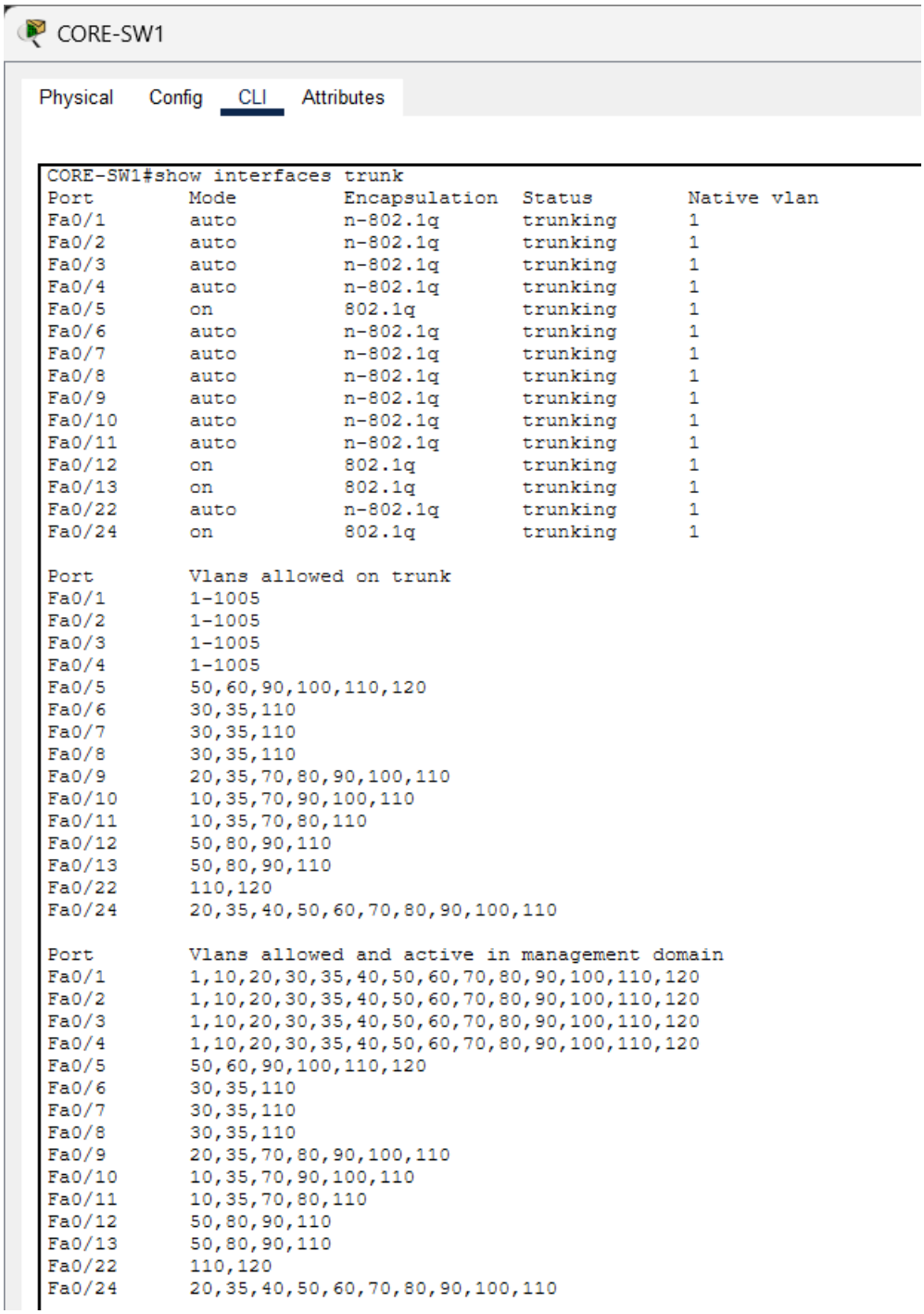
5.3 Addressing Rationale

VLSM was used at the main campus to allocate address space according to the expected number of devices in each network while minimizing address wastage. High-capacity networks, particularly the student residence and student wireless networks, were assigned larger address blocks to accommodate large numbers of concurrent users. Smaller networks, including servers, printers, voice, management and ICT administration, were allocated smaller subnets because they require fewer host addresses. Each VLAN therefore forms a separate IP subnet with its own HSRP virtual default gateway. The satellite campus uses /24 networks to provide a simple and consistent addressing structure with sufficient capacity for future expansion. Separating the address space by function also simplifies routing, ACL configuration, fault isolation, monitoring and network administration.

6 Routing Redundancy and Layer 2 Resilience

6.1 Inter VLAN Routing

The main-campus core switches use switched virtual interfaces (SVIs) and Layer 3 IP routing to route traffic between VLANs. Access switches carry the required VLANs to the core over IEEE 802.1Q trunk links. HSRP virtual IP addresses configured on the main-campus SVIs provide the default gateways used by end devices. At the satellite campus, SAT-CORE-SW performs inter-VLAN routing for the satellite VLANs using local SVI gateway addresses. DHCP relay addresses configured on the relevant SVIs forward client DHCP requests to the appropriate DHCP service.



```
Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20,30,35,40,50,60,70,80,90,100,110,120
Fa0/2       1,10,20,30,35,40,50,60,70,80,90,100,110,120
Fa0/3       1,10,20,30,35,40,50,60,70,80,90,100,110,120
Fa0/4       1,10,20,30,35,40,50,60,70,80,90,100,110,120
Fa0/5       50,60,90,100,110,120
Fa0/6       30,35,110
Fa0/7       30,35,110
Fa0/8       30,35,110
Fa0/9       20,35,70,80,90,100,110
Fa0/10      10,35,70,90,100,110
Fa0/11      10,35,70,80,110
Fa0/12      50,80,90,110
Fa0/13      50,80,90,110
Fa0/22      110,120
Fa0/24      20,35,40,50,60,70,80,90,100,110

CORE-SW1#
```

Figure 6.1: IEEE 802.1Q trunk configuration and VLAN forwarding status on CORE-SW1.

6.2 HSRP

HSRP provides a stable virtual default gateway on the main campus. CORE-SW1 normally operates as Active with priority 110 and preemption enabled, while CORE-SW2 operates as Standby with priority 100. Normal Active and Standby operation was verified across all configured VLAN groups, including Admin, Staff, Research, Printers, Students, Residence, Servers, Voice, IoT, Wireless, Management and ICT administration. A controlled failover test was performed on VLAN 10 by temporarily reducing CORE-SW1's HSRP priority below that of CORE-SW2. CORE-SW2 then became Active for the VLAN 10 virtual gateway. After CORE-SW1's priority was restored to 110, preemption returned the Active role to CORE-SW1.

```
CORE-SW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   110 P Active   local           10.10.7.3       10.10.7.1
Vl120       20   110 P Active   local           10.10.4.131     10.10.4.129
Vl130       30   110 P Active   local           10.10.7.131     10.10.7.129
Vl135       35   110 P Active   local           10.10.6.3       10.10.6.1
Vl140       40   110 P Active   local           10.10.4.3       10.10.4.1
Vl150       50   110 P Active   local           10.10.8.3       10.10.8.1
Vl160       60   110 P Active   local           10.10.6.195     10.10.6.193
Vl170       70   110 P Active   local           10.10.6.67      10.10.6.65
Vl180       80   110 P Active   local           10.10.5.131     10.10.5.129
Vl190       90   110 P Active   local           10.10.5.3       10.10.5.1
Vl1100      100  110 P Active   local           10.10.0.3       10.10.0.1
Vl1110      110  110 P Active   local           10.10.6.131     10.10.6.129
Vl1120      120  110 P Active   local           10.10.12.3      10.10.12.1
CORE-SW1#
```

Figure 6.2(a): HSRP status on CORE-SW1 showing Active gateway roles and Standby peers across the main-campus VLANs.

```
CORE-SW2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   100 P Standby  10.10.7.2       local           10.10.7.1
Vl20        20   100 P Standby  10.10.4.130     local           10.10.4.129
Vl30        30   100 P Standby  10.10.7.130     local           10.10.7.129
Vl35        35   100 P Standby  10.10.6.2       local           10.10.6.1
Vl40        40   100 P Standby  10.10.4.2       local           10.10.4.1
Vl50        50   100 P Standby  10.10.8.2       local           10.10.8.1
Vl60        60   100 P Standby  10.10.6.194     local           10.10.6.193
Vl70        70   100 P Standby  10.10.6.66      local           10.10.6.65
Vl80        80   100 P Standby  10.10.5.130     local           10.10.5.129
Vl90        90   100 P Standby  10.10.5.2       local           10.10.5.1
Vl100       100  100 P Standby  10.10.0.2       local           10.10.0.1
Vl110       110  100 P Standby  10.10.6.130     local           10.10.6.129
Vl120       120  100 P Standby  10.10.12.2      local           10.10.12.1
CORE-SW2#
```

Figure 6.2(b): HSRP status on CORE-SW2 showing the Standby gateway roles.

```
CORE-SW1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
CORE-SW1(config)#interface vlan 10
CORE-SW1(config-if)#standby 10 priority 90
CORE-SW1(config-if)#end
CORE-SW1#
*Sep 30, 13:50:19.5050: SYS-5-CONFIG_I: Configured from console by console
CORE-SW1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Vl10        10   90  P Speak    10.10.7.3       10.10.7.3       10.10.7.1
Vl20        20   110 P Active   local           10.10.4.131     10.10.4.129
Vl30        30   110 P Active   local           10.10.7.131     10.10.7.129
Vl35        35   110 P Active   local           10.10.6.3       10.10.6.1
Vl40        40   110 P Active   local           10.10.4.3       10.10.4.1
Vl50        50   110 P Active   local           10.10.8.3       10.10.8.1
Vl60        60   110 P Active   local           10.10.6.195     10.10.6.193
Vl70        70   110 P Active   local           10.10.6.67      10.10.6.65
Vl80        80   110 P Active   local           10.10.5.131     10.10.5.129
Vl90        90   110 P Active   local           10.10.5.3       10.10.5.1
Vl100       100  110 P Active   local           10.10.0.3       10.10.0.1
Vl110       110  110 P Active   local           10.10.6.131     10.10.6.129
Vl120       120  110 P Active   local           10.10.12.3      10.10.12.1
CORE-SW1#
```

Figure 6.2(c): Controlled HSRP failover test on VLAN 10 after reducing CORE-SW1's priority to 90, causing CORE-SW2 (10.10.7.3) to assume the Active gateway role.

6.3 Spanning Tree and Redundant Uplinks

Per-VLAN Spanning Tree (PVST) prevents Layer 2 switching loops created by the redundant uplinks between the access and core switches. CORE-SW1 was assigned a lower spanning-tree bridge priority for the required VLANs, making it the preferred root bridge, while CORE-SW2 provides an alternative redundant path. During verification, primary uplinks operated in the Forwarding state while redundant paths entered an Alternate/Blocking state where required. If a forwarding path becomes unavailable, spanning tree can reconverge and activate an alternative path. PortFast is enabled on appropriate end-device access ports to allow them to transition rapidly to the forwarding state. Lightweight access point (LAP) and WLC connections use the required trunk configurations.

```
CORE-SW1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID     Priority    32778
              Address     0007.EC89.BEE8
              This bridge is the root
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32778  (priority 32768 sys-id-ext 10)
              Address     0007.EC89.BEE8
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/4            Desg FWD 19        128.4    P2p
Fa0/10           Desg FWD 19        128.10   P2p
Fa0/11           Desg FWD 19        128.11   P2p

CORE-SW1#
```

Figure 6.3(a): PVST verification showing CORE-SW1 as the root bridge for VLAN 10.

```
CORE-SW2#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol ieee
  Root ID     Priority    32778
              Address     0007.EC89.BEE8
              Cost        38
              Port        10(FastEthernet0/10)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID   Priority    32778  (priority 32768 sys-id-ext 10)
              Address     0090.2B16.9801
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- --------------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/2            Desg FWD 19        128.2    P2p
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/10           Root FWD 19        128.10   P2p
Fa0/11           Altn BLK 19        128.11   P2p
Fa0/4            Desg FWD 19        128.4    P2p

CORE-SW2#
```

Figure 6.3(b): PVST verification on CORE-SW2 showing the forwarding root port and blocked alternate path for VLAN 10.

7 Security Architecture

7.1 Segmentation and Zero Trust Policy

The network applies least-privilege principles by placing different user and device categories in separate VLANs and enforcing extended ACLs at Layer 3 boundaries. Student and residence networks are prevented from initiating unrestricted access to protected internal campus networks. Research-laboratory users can access authorized central services but are isolated from unrelated user and management networks. IoT devices are restricted to required registration, management and service

communication, while ICT administration receives broader access for authorized network-management and support tasks.

Control	Implementation and purpose
MAIN-STUDENT-ZERO-TRUST	Allows DHCP, DNS and approved services, then denies protected campus ranges before permitting external traffic
RESEARCH-POLICY	Allows central DNS and approved server access while denying lateral movement into user and management subnets
SERVER-ZERO-TRUST	Allows authorized management, RADIUS, IoT, NTP, syslog and service flows; denies other access to the server network
SAT-STUDENT-POLICY	Restricts satellite students while allowing required DNS and approved services
Zone-based firewall	Inspects authorized inside-to-outside sessions and drops unsolicited default traffic

```
CORE-SW1#show access-lists
Extended IP access list STUDENT-POLICY
    10 permit ip any host 10.10.6.10
    20 permit udp any host 10.10.6.200 eq domain
    30 permit tcp any host 10.10.6.200 eq domain
    40 deny ip any 10.10.6.192 0.0.0.63
    50 deny ip any 10.10.4.128 0.0.0.127
    60 deny ip any 10.10.5.0 0.0.0.127
    70 deny ip any 10.10.5.128 0.0.0.127
    80 deny ip any 10.10.6.0 0.0.0.63
    90 deny ip any 10.10.6.64 0.0.0.63
    100 deny ip any 10.10.6.128 0.0.0.63
    110 permit ip any any
Extended IP access list RESEARCH-POLICY
    10 permit udp any host 10.10.6.200 eq domain
    20 permit tcp any host 10.10.6.200 eq domain
    30 permit ip any 10.10.6.0 0.0.0.63
    40 deny ip any 10.10.7.0 0.0.0.127
    50 deny ip any 10.10.4.128 0.0.0.127
    60 deny ip any 10.10.5.0 0.0.0.127
    70 deny ip any 10.10.5.128 0.0.0.127
    80 deny ip any 10.10.6.192 0.0.0.63
    90 deny ip any 10.10.6.64 0.0.0.63
    100 deny ip any 10.10.6.128 0.0.0.63
    110 permit ip any any (300 match(es))
Extended IP access list RESIDENCE-POLICY
    10 permit udp any host 10.10.6.200 eq domain
    20 permit tcp any host 10.10.6.200 eq domain
    30 deny ip any 10.10.7.0 0.0.0.127
    40 deny ip any 10.10.7.128 0.0.0.127
    50 deny ip any 10.10.4.128 0.0.0.127
    60 deny ip any 10.10.4.0 0.0.0.127
    70 deny ip any 10.10.5.0 0.0.0.127
    80 deny ip any 10.10.5.128 0.0.0.127
    90 deny ip any 10.10.6.0 0.0.0.63
    100 deny ip any 10.10.6.64 0.0.0.63
    110 deny ip any 10.10.6.128 0.0.0.63
    120 deny ip any 10.10.6.192 0.0.0.63
    130 permit ip any any
Standard IP access list ICT-MANAGEMENT-ONLY
    10 permit 10.10.12.0 0.0.0.31
    20 permit host 10.10.6.204
    30 deny any
Extended IP access list SERVER-ZERO-TRUST
    10 permit ip 10.10.12.0 0.0.0.31 10.10.6.192 0.0.0.63 (148 match(es))
    20 permit udp any host 10.10.6.200 eq domain
    30 permit tcp any host 10.10.6.200 eq domain
    40 permit udp any host 10.10.6.200 eq bootps
    50 permit udp 10.10.6.128 0.0.0.63 host 10.10.6.201 eq 1645 (14 match(es))
    60 permit udp 10.10.6.128 0.0.0.63 host 10.10.6.201 eq 1812
    70 permit ip 10.10.5.128 0.0.0.127 host 10.10.6.202 (5502 match(es))
```

```
    80 permit ip 10.20.80.0 0.0.0.255 host 10.10.6.202 (714 match(es))
    90 permit udp any host 10.10.6.203 eq 514 (10 match(es))
    100 permit udp any host 10.10.6.203 eq 123 (877 match(es))
    110 permit ip host 10.10.6.153 host 10.10.6.204 (393 match(es))
    120 permit udp host 10.10.5.10 host 10.10.6.201 eq 1645 (18 match(es))
    130 permit udp host 10.10.5.10 host 10.10.6.201 eq 1812
    140 permit udp host 10.20.90.10 host 10.10.6.201 eq 1645 (3 match(es))
    150 permit udp host 10.20.90.10 host 10.10.6.201 eq 1812
    160 deny ip any 10.10.6.192 0.0.0.63
    170 permit ip any any
Extended IP access list MAIN-IOT-ZERO-TRUST
    10 permit udp any any eq bootps (53 match(es))
    20 permit udp any host 10.10.6.200 eq domain
    30 permit tcp any host 10.10.6.200 eq domain
    40 permit ip any host 10.10.6.202 (5502 match(es))
    50 permit udp any host 224.0.0.2 eq 1985 (283 match(es))
    60 permit udp any host 224.0.0.102 eq 1985
    70 deny ip any any (3613 match(es))
Extended IP access list MAIN-STUDENT-ZERO-TRUST
    10 permit udp any any eq bootps (67 match(es))
    20 permit udp any host 10.10.6.200 eq domain
    30 permit tcp any host 10.10.6.200 eq domain
    40 permit ip any 10.10.6.0 0.0.0.63
    50 permit icmp any host 10.10.4.1
    60 permit icmp any host 10.10.8.1
    70 permit icmp any host 10.10.0.1
    80 permit ip any host 10.10.5.10 (35 match(es))
    90 deny ip any 10.10.0.0 0.0.255.255
    100 deny ip any 10.20.0.0 0.0.255.255
    110 permit ip any any (907 match(es))

CORE-SW1#
```

Figure 7.1(a): Extended ACL verification on CORE-SW1 showing zero-trust segmentation policies and traffic match counters.

```
MAIN-EDGE-RTR>enable
Password:
MAIN-EDGE-RTR#show zone security
zone self
  Description: System defined zone


zone INSIDE
  Member Interfaces:
    FastEthernet0/0
    FastEthernet0/1


zone OUTSIDE
  Member Interfaces:
    FastEthernet1/0


MAIN-EDGE-RTR#show zone-pair security
Zone-pair name MAIN-TO-OUTSIDE
    Source-Zone INSIDE  Destination-Zone OUTSIDE
    service-policy MAIN-TO-OUTSIDE-POLICY

Zone-pair name OUTSIDE-TO-MAIN
    Source-Zone OUTSIDE  Destination-Zone INSIDE
    service-policy OUTSIDE-TO-MAIN-POLICY

MAIN-EDGE-RTR#
```

Figure 7.1(b): Zone-Based Firewall configuration on MAIN-EDGE-RTR showing the INSIDE and OUTSIDE security zones and associated zone pairs.

MAIN-EDGE-RTR

Physical Config CLI Attributes

```
class-map type inspect match-all MAIN-INSIDE-CLASS
 match access-group name MAIN-INSIDE-TRAFFIC
class-map type inspect match-all SATELLITE-VPN-CLASS
 match access-group name SATELLITE-VPN-IN
class-map match-any MAIN-VOICE-CLASS
 match access-group name MAIN-VOICE-QOS
 match ip dscp ef
class-map match-any MAIN-VIDEO-CLASS
 match access-group name MAIN-VIDEO-QOS
!
policy-map type inspect MAIN-TO-OUTSIDE-POLICY
 class type inspect MAIN-INSIDE-CLASS
  inspect
 class type inspect class-default
  drop
!
policy-map type inspect OUTSIDE-TO-MAIN-POLICY
 class type inspect SATELLITE-VPN-CLASS
  pass
 class type inspect class-default
  drop
```

Figure 7.1(c): Zone-Based Firewall policies on MAIN-EDGE-RTR showing inspected inside-to-outside traffic, permitted satellite VPN traffic and default traffic dropping.

7.2 Access Layer Hardening

Access-layer switch hardening includes port security on appropriate endpoint-facing interfaces, PortFast on suitable access ports, explicit access VLAN assignments and restricted trunk VLAN lists. Unused interfaces are administratively shut down to reduce the available attack surface. Remote switch management is restricted to authorized ICT administration addresses and uses SSH rather than insecure remote-access protocols.

```
ADMIN-SW2>enable
Password:
Password:
ADMIN-SW2#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)         (Count)
---------------------------------------------------------------------
        Fa0/1        3           2                0         Restrict
        Fa0/2        3           2                0         Restrict
        Fa0/3        3           2                0         Restrict
        Fa0/4        1           1                0         Restrict
        Fa0/5       20           4                0         Restrict
---------------------------------------------------------------------
ADMIN-SW2#
```

Figure 7.2(a): Port-security verification on ADMIN-SW2 showing secure access ports, MAC address limits and restrict violation mode.

```
ADMIN-SW2#show port-security interface fa0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 3
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 000B.BE84.C88A:70
Security Violation Count   : 0

ADMIN-SW2#
```

Figure 7.2(b): Detailed port-security status for ADMIN-SW2 FastEthernet0/1.

7.3 Central Authentication

The RADIUS server at 10.10.6.201 provides centralized 802.1X authentication. Infrastructure devices use the configured shared secret and authorized management source network to communicate with the authentication service. Wired research-laboratory ports were configured as 802.1X authenticators with global 802.1X control enabled. Wireless staff and student WLANs use enterprise authentication through the WLC. Testing confirmed that Packet Tracer clients require the 802.1X security option to be enabled and valid credentials to be supplied before authenticated network access succeeds.

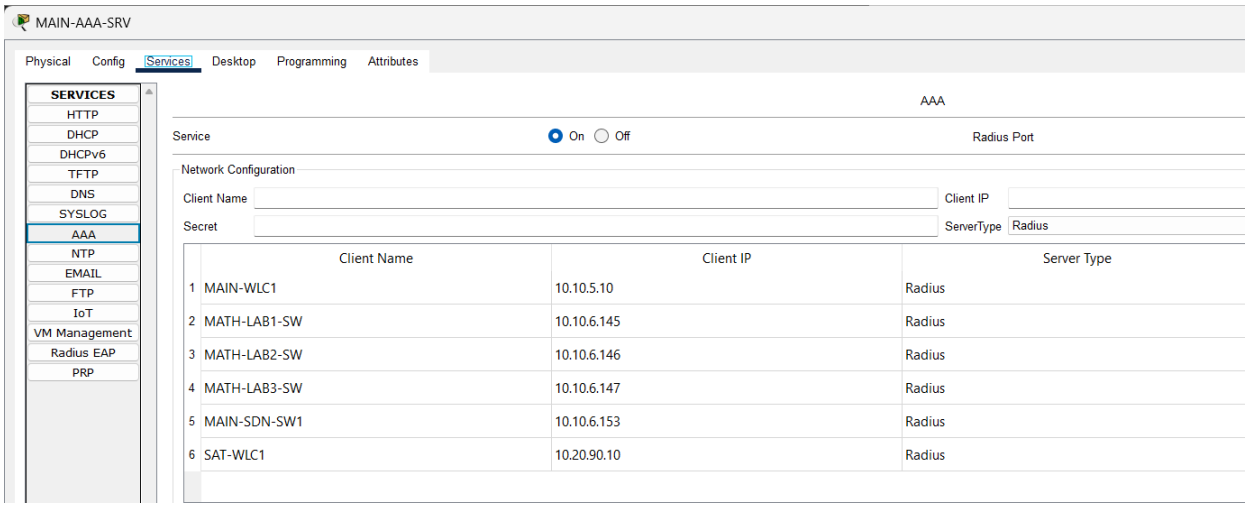


Figure 7.3(a): Central RADIUS/AAA configuration showing authorised network clients.

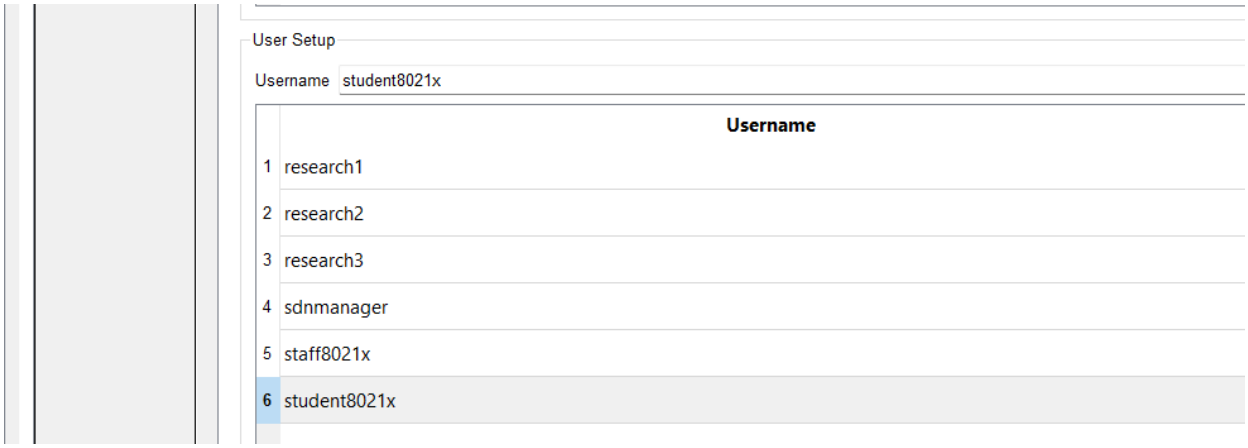


Figure 7.3(b): RADIUS user accounts configured for 802.1X authentication.

7.4 Security Testing

Test	Expected result	Observed result	Status
Unauthorized student-to-protected network	Traffic denied by inbound ACL	Restricted traffic blocked; required DHCP/DNS retained	Pass
Unused switch port	No link or network access	Port remained administratively down	Pass
802.1X with incorrect credentials	Association or authorization denied	Client did not obtain authorized access	Pass

802.1X with valid credentials	Authentication and correct VLAN service	Client connected and received DHCP service	Pass
VPN inspection	Active security associations and increasing counters	IKE SA reached QM_IDLE and IPsec packet counters increased during protected inter-campus traffic.	Pass

8 Wireless and 802.1X Implementation

8.1 Wireless Architecture

Lightweight access points (LAPs) were deployed in the library, mathematics building, lecture halls, residences, ICT/server area and satellite campus. Each LAP joins its local wireless LAN controller (WLC) through the configured management network. User WLAN traffic is mapped to the appropriate staff, student or residence VLAN. Trunk links between the WLCs, LAP access switches and the switching infrastructure carry only the VLANs required at each location. Wireless access is centrally managed by the WLCs, while RADIUS provides 802.1X authentication for the protected WLANs.

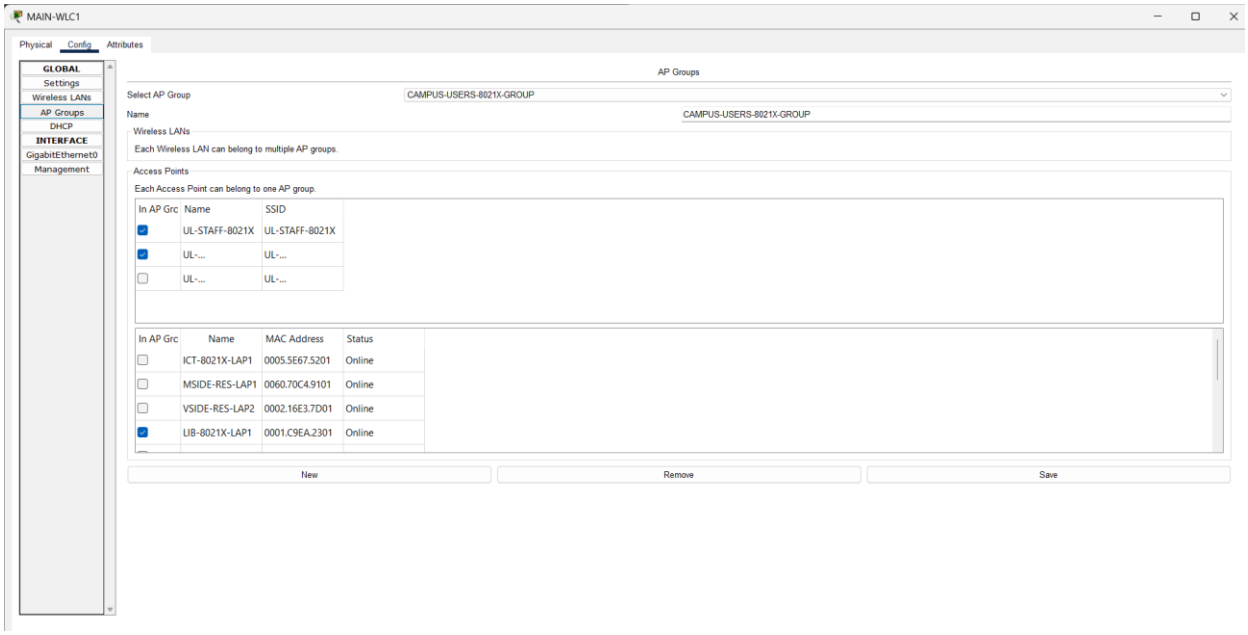


Figure 8.1: MAIN-WLC1 campus 802.1X AP group showing registered lightweight access points and WLAN assignments.

Wireless service	SSID	VLAN	Authentication
Main staff	UL-STAFF-8021X	90	802.1X through RADIUS
Main students	UL-STUDENT-8021X	100	802.1X through RADIUS
Residence students	UL-RESIDENCE-8021X	50	802.1X through RADIUS
Satellite staff	UL-STAFF-8021X	90	802.1X through RADIUS
Satellite students	UL-STUDENT-8021X	100	802.1X through RADIUS

The main and satellite WLANs use the group name CAMPUS-USERS-8021X-GROUP to maintain consistent authentication administration. Staff and student user accounts are centrally configured on the RADIUS service, allowing the wireless LAN controllers to authenticate users before granting access to the appropriate WLAN and VLAN.

8.2 Residence Wireless Design

Residence users are separated from the general student wireless network because the residence network has different capacity and security requirements. Two lightweight access points (LAPs) were deployed on each residence access switch to improve wireless coverage and support higher client density. The residence WLAN is mapped to VLAN 50, while the LAP uplinks and associated switch trunks carry the management and wireless VLANs required by the deployed configuration. The native VLAN is aligned with the LAP management design, allowing the access points to communicate correctly with the wireless LAN controller while residence client traffic is carried on its assigned VLAN.

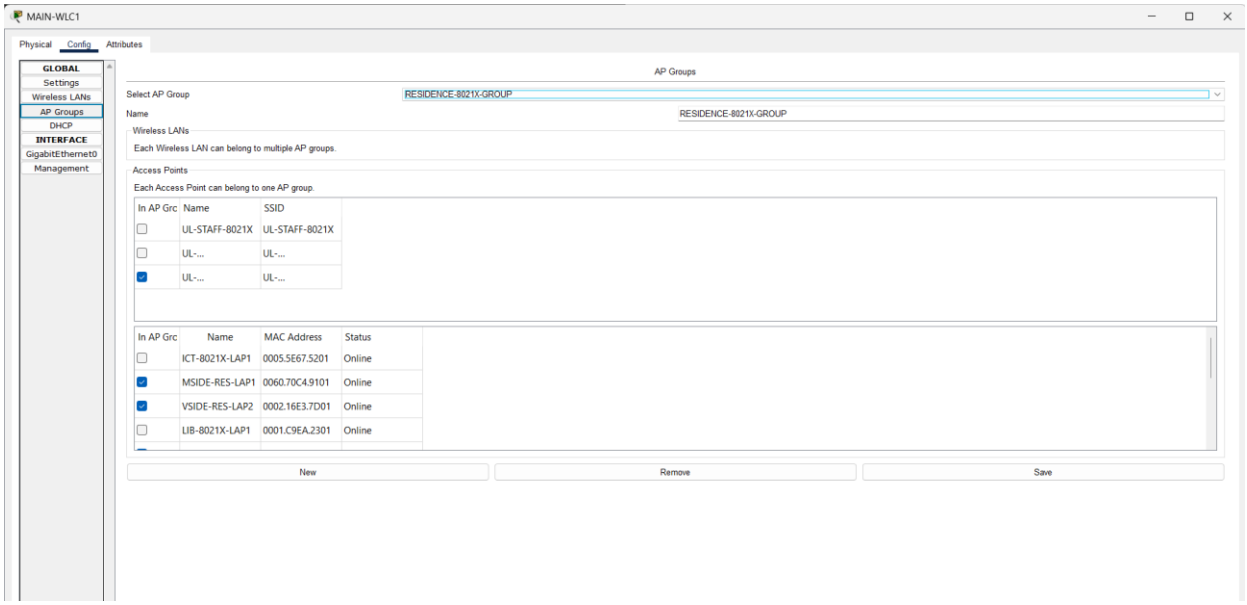


Figure 8.2(a): MAIN-WLC1 residence 802.1X AP group showing online residence lightweight access points.



Figure 8.2(b): Successful WPA2-Enterprise residence wireless connection showing DHCP addressing, gateway and central DNS configuration.

9 VPN and Multi Site Connectivity

MAIN-EDGE-RTR and SAT-EDGE-RTR establish a site-to-site IPsec VPN across the simulated ISP, providing secure communication between the main-campus network 10.10.0.0/16 and the satellite-campus network 10.20.0.0/16. IKE negotiation successfully reached the QM_IDLE state, confirming establishment of the IKE security association. IPsec verification showed active inbound and outbound ESP security associations, while encapsulation and decapsulation counters increased as protected traffic was exchanged between the two campuses. This confirmed successful bidirectional encrypted communication through the VPN tunnel.

VPN component	Main campus	Satellite campus
Public endpoint	203.0.113.1	198.51.100.2

Protected network	10.10.0.0/16	10.20.0.0/16
Crypto map	UL-VPN	UL-VPN
Observed state	QM_IDLE / ACTIVE	QM_IDLE / ACTIVE
Traffic evidence	Encapsulation and decapsulation counters increased	Encapsulation and decapsulation counters increased

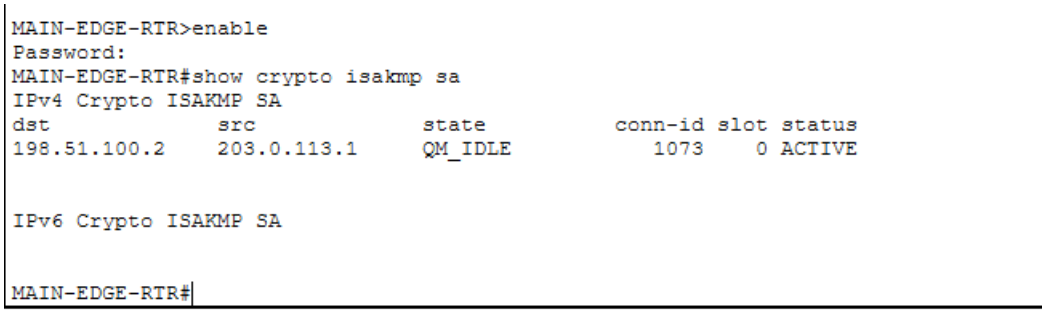


Figure 9.1(a): IKE security association on MAIN-EDGE-RTR showing the site-to-site VPN in QM_IDLE and ACTIVE state.

```
MAIN-EDGE-RTR#show crypto ipsec sa

interface: FastEthernet1/0
    Crypto map tag: UL-VPN, local addr 203.0.113.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
   remote  ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
   current_peer 198.51.100.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 2414, #pkts encrypt: 2414, #pkts digest: 0
   #pkts decaps: 2433, #pkts decrypt: 2433, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 203.0.113.1, remote crypto endpt.:198.51.100.2
     path mtu 1500, ip mtu 1500, ip mtu idb FastEthernet1/0
     current outbound spi: 0x135D49DC(324880860)

     inbound esp sas:
      spi: 0x6E39D110(1849282832)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2002, flow_id: FPGA:1, crypto map: UL-VPN
        sa timing: remaining key lifetime (k/sec): (4525504/1370)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0x135D49DC(324880860)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2003, flow_id: FPGA:1, crypto map: UL-VPN
        sa timing: remaining key lifetime (k/sec): (4525504/1370)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE

     outbound ah sas:

     outbound pcp sas:

MAIN-EDGE-RTR#
```

Figure 9.1(b): IPsec security association on MAIN-EDGE-RTR showing active encrypted traffic between the main and satellite campus networks.

10 VoIP and Quality of Service

10.1 Voice Services

Cisco CallManager Express (CME) was configured on MAIN-VOICE-RTR and SAT-VOICE-RTR to provide call-control services at the two campuses. Main-campus IP phones use extensions in the 2000 range, while satellite-campus phones use extensions in the 3000 range. DHCP option 150 directs IP phones to their respective local call-control addresses, allowing the phones to register automatically with the appropriate CME router. Registered ephones display their assigned extensions and remain reachable through their local voice VLANs. Inter-site dial peers route 3xxx calls from the main campus to the satellite voice gateway and 2xxx calls from the satellite campus to the main voice gateway. A successful

call between extensions 2001 and 3001 confirmed inter-campus dial-plan operation and voice reachability across the multi-site network.

Site	Call-control address	Extension range	Inter-site dial peer
Main campus	10.10.6.68:2000	2001-2009	3... to 10.20.70.2
Satellite campus	10.20.70.2	3001 onward	2... to 10.10.6.68

```
Password:
MAIN-VOICE-RTR#show ephone

ephone-1 Mac:00E0.F740.D2D2 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.80 1025 7960   keepalive 43 max_line 2
 button 1: dn 1  number 2001 CH1   IDLE

ephone-2 Mac:0060.2F20.7935 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.76 1025 7960   keepalive 43 max_line 2
 button 1: dn 5  number 2005 CH1   IDLE

ephone-3 Mac:0001.96BA.D57C TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.81 1025 7960   keepalive 43 max_line 2
 button 1: dn 4  number 2004 CH1   IDLE

ephone-4 Mac:00D0.D3AC.6BD8 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.79 1025 7960   keepalive 43 max_line 2
 button 1: dn 2  number 2002 CH1   IDLE

ephone-5 Mac:000A.F399.1A92 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.77 1025 7960   keepalive 43 max_line 2
 button 1: dn 6  number 2006 CH1   IDLE

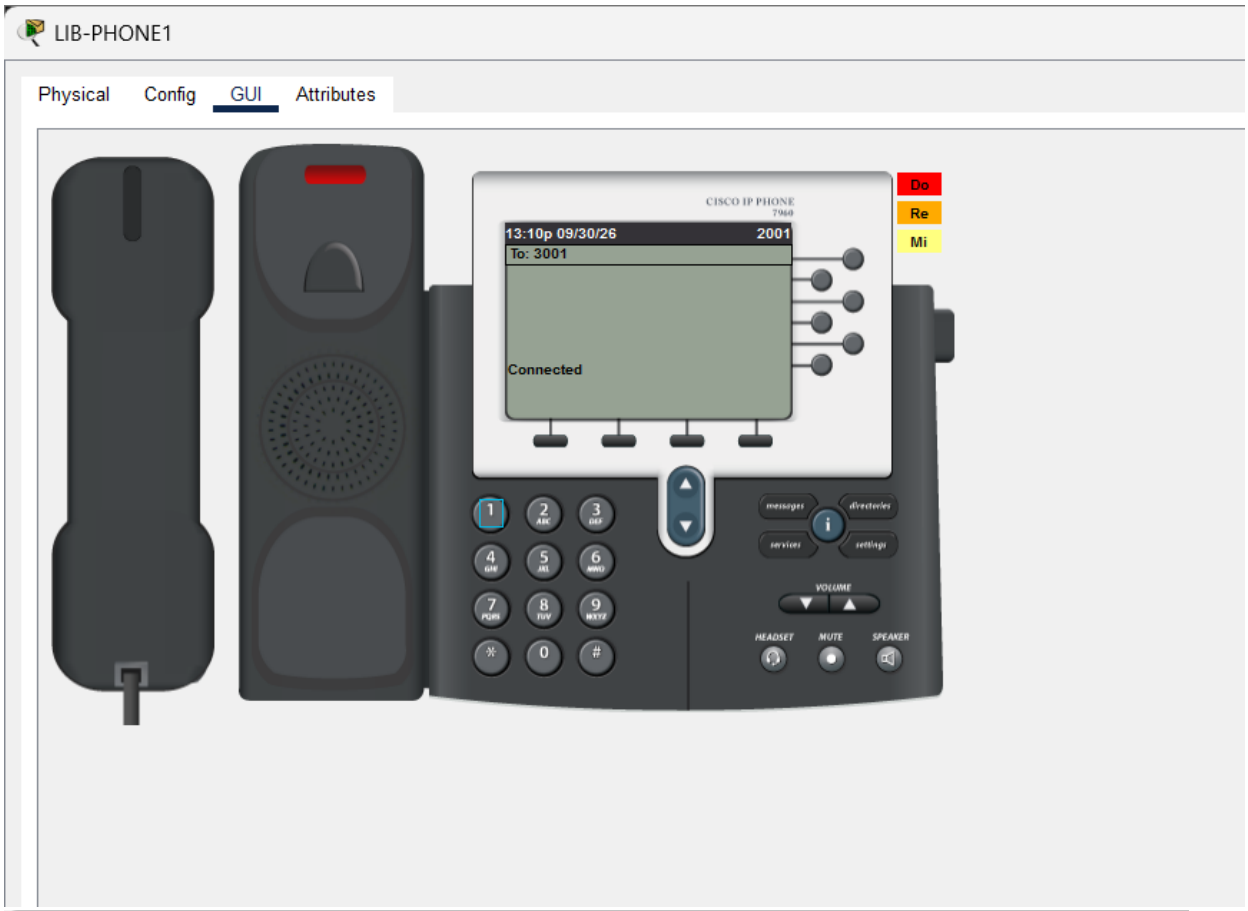
ephone-6 Mac:0001.9603.3D66 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.82 1025 7960   keepalive 43 max_line 2
 button 1: dn 3  number 2003 CH1   IDLE

ephone-7 Mac:000B.BE84.C88A TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.83 1025 7960   keepalive 43 max_line 2
 button 1: dn 9  number 2009 CH1   IDLE

ephone-8 Mac:0001.C987.A416 TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.84 1025 7960   keepalive 43 max_line 2
 button 1: dn 7  number 2007 CH1   IDLE

ephone-9 Mac:0030.A30C.C04A TCP socket:[1] activeLine:0 REGISTERED in SCCP ver 12 and Server in ver 8
mediaActive:0 offhook:0 ringing:0 reset:0 reset_sent:0 paging 0 debug:0 caps:8
IP:10.10.6.78 1025 7960   keepalive 43 max_line 2
 button 1: dn 8  number 2008 CH1   IDLE
MAIN-VOICE-RTR#
```

Figure 10.1(a): MAIN-VOICE-RTR ephone status showing registered main-campus IP phones and assigned 2000-series extensions.


LIB-PHONE1
Physical
Config
GUI
Attributes
CISCO IP PHONE
7960
13:10p 09/30/26
2001
To: 3001
Connected
Do
Re
Mi
messages
directories
services
settings
VOLUME
HEADSET
MUTE
SPEAKER


SAT-PHONE1
Physical
Config
GUI
Attributes
CISCO IP PHONE
7960
13:10p 09/30/26
3001
From: 2001
Connected
Do
Re
Mi
messages
directories
services
settings
VOLUME
HEADSET
MUTE
SPEAKER

Figure 10.1(b): Successful inter-campus VoIP call between main-campus extension 2001 and satellite extension 3001.

10.2 QoS Design

WAN Quality of Service (QoS) policies are applied outbound on the WAN-facing interfaces of MAIN-EDGE-RTR and SAT-EDGE-RTR. Voice traffic is assigned a strict-priority queue equal to 20% of the available interface bandwidth. On the main-campus edge router, video traffic is allocated 30% of the interface bandwidth, while remaining traffic uses the default class. Voice traffic is identified and marked with DSCP EF before encryption so that delay-sensitive voice communication receives preferential treatment across the WAN.

Router	Policy	Voice treatment	Additional treatment
MAIN-EDGE-RTR	MAIN-WAN-QOS	Strict priority 20%	Video bandwidth 30%
SAT-EDGE-RTR	SAT-WAN-QOS	Strict priority 20%	Fair queue for default traffic

```
MAIN-EDGE-RTR>enable
Password:
MAIN-EDGE-RTR#show ephone
MAIN-EDGE-RTR#show ephone
MAIN-EDGE-RTR#show policy-map interface FastEthernet1/0
 FastEthernet1/0

  Service-policy output: MAIN-WAN-QOS

    Class-map: MAIN-VOICE-CLASS (match-any)
      1 packets, 30 bytes
      5 minute offered rate 1 bps, drop rate 0 bps
      Match: access-group name MAIN-VOICE-QOS
        1 packets, 30 bytes
        5 minute rate 1 bps
      Match: ip dscp ef (46)
        0 packets, 0 bytes
        5 minute rate 0 bps
      Queueing
        Strict Priority
        Output Queue: Conversation 264
        Bandwidth 20 (%)
        Bandwidth 20000 (kbps) Burst 500000 (Bytes)
        (pkts matched/bytes matched) 0/0
        (total drops/bytes drops) 0/0

    Class-map: MAIN-VIDEO-CLASS (match-any)
      0 packets, 0 bytes
      5 minute offered rate 0 bps, drop rate 0 bps
      Match: access-group name MAIN-VIDEO-QOS
        0 packets, 0 bytes
        5 minute rate 0 bps
      Queueing
        Output Queue: Conversation 265
        Bandwidth 30 (%)
        Bandwidth 30000 (kbps)Max Threshold 64 (packets)
        (pkts matched/bytes matched) 0/0
        (depth/total drops/no-buffer drops) 0/0/0

    Class-map: class-default (match-any)
      144 packets, 16626 bytes
      5 minute offered rate 402 bps, drop rate 0 bps
      Match: any
      Queueing
        Flow Based Fair Queueing
        Maximum number of Hashed Queues 256
        Bandwidth 75000 (kbps)Max Threshold 64 (packets)
        (total queued/total drops/no-buffer drops) 0/0/0

MAIN-EDGE-RTR#
```

Figure 10.2: QoS service policy on MAIN-EDGE-RTR showing DSCP EF voice classification, 20% strict-priority bandwidth and 30% video bandwidth allocation.

11 IoT and Central Services

11.1 Central Services

Centralized network services are hosted at the main campus to support addressing, name resolution, authentication, IoT management, monitoring and network programmability. These services are assigned dedicated addresses and are separated from general user networks through the campus VLAN and security architecture.

Service	Address	Function
DNS and DHCP	10.10.6.200	Name resolution and central address allocation
RADIUS	10.10.6.201	802.1X authentication
IoT registration	10.10.6.202	Registers and manages smart devices
NTP and Syslog	10.10.6.203	Time synchronization and central logging
SDN controller	10.10.6.204	Network discovery and programmable monitoring
Main WLC	10.10.5.10	Main-campus wireless control

11.2 IoT Segmentation

Cameras, smoke detectors, temperature sensors and related smart devices are placed in dedicated IoT VLANs to separate them from general user traffic. Main-campus IoT devices use VLAN 80 and the 10.10.5.128/25 network, while satellite-campus IoT devices use the 10.20.80.0/24 network. ACLs restrict communication between the IoT networks and unrelated campus networks while permitting the traffic required for device registration and management through the central IoT service at 10.10.6.202.





Figure 11.1: Central IoT registration server showing registered cameras, environmental sensors and smart devices from the main and satellite campuses.

11.3 Monitoring

Network infrastructure devices synchronize their clocks with the central NTP service and forward supported log events to the central syslog server. Consistent timestamps support event correlation, incident investigation and configuration auditing. The SDN monitoring application communicates with the controller through its API to obtain information about managed network devices, including hostname, management address, device type, reachability and collection status.

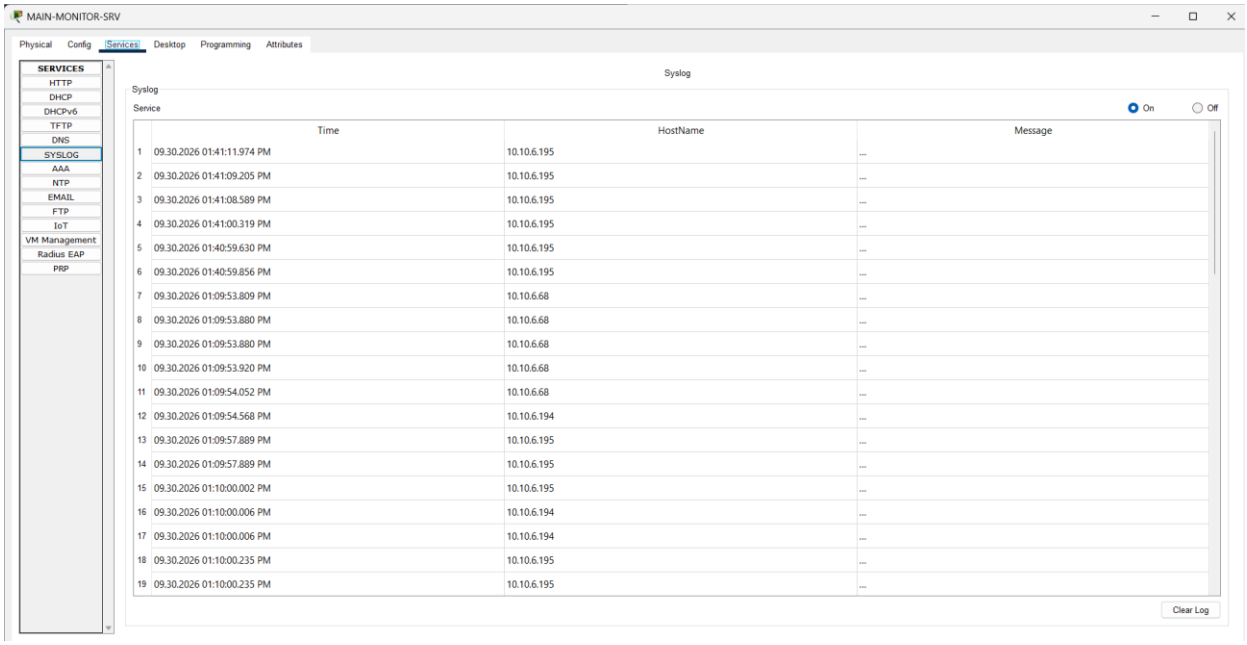


Figure 11.3(a): Central Syslog service on MAIN-MONITOR-SRV showing timestamped log entries received from multiple network devices.

```
CORE-SW1>enable
Password:
CORE-SW1#show clock
13:43:3.178 UTC Wed Sep 30 2026
CORE-SW1#show ntp associations

address         ref clock       st   when     poll    reach   delay          offset            disp
 ~10.10.6.203   127.127.1.1     1    27       16      376     4218.00        2110.00           0.13
 * sys.peer, # selected, + candidate, - outlyer, x falseticker, ~ configured
CORE-SW1#
```

Figure 11.3(b): CORE-SW1 NTP association with the central monitoring server at 10.10.6.203.

12 SDN Integration and Automation

12.1 SDN Controller

The SDN controller at 10.10.6.204 is located in the protected server network. The ICT administration workstation at 10.10.12.11 can reach the controller through the authorized management path. The controller provides centralized visibility of managed network infrastructure and exposes REST API endpoints that support network monitoring and programmability.

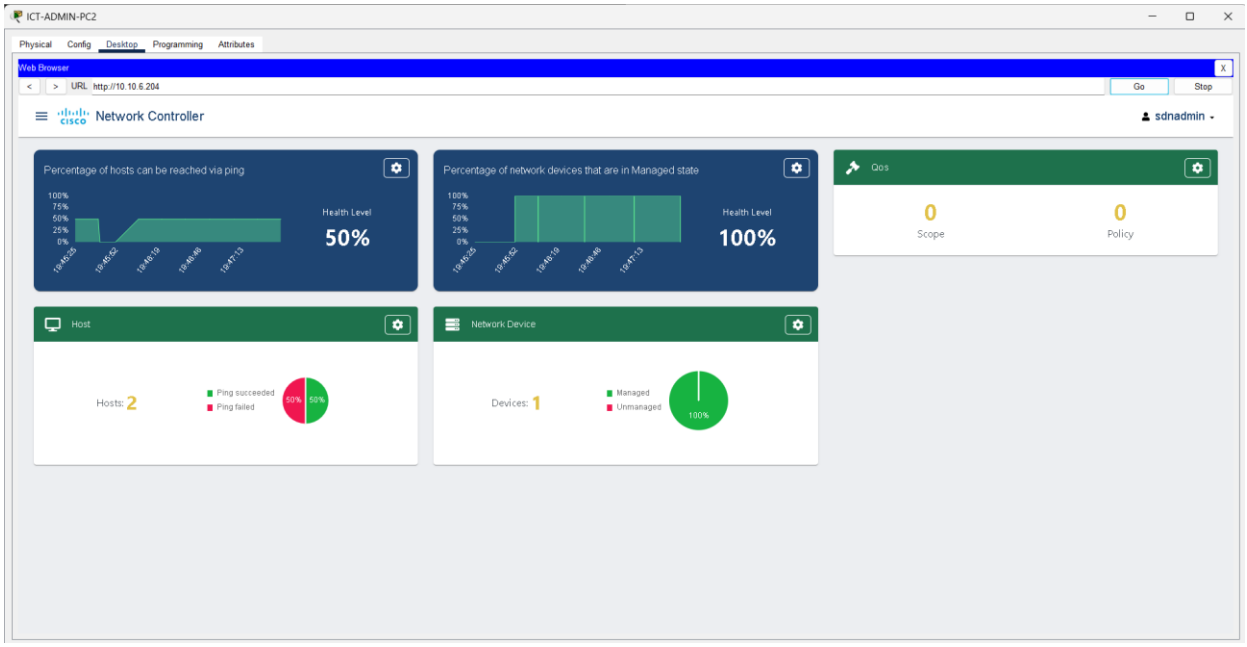


Figure 12.1: SDN controller dashboard showing centralised monitoring and a managed network device.

12.2 Monitoring Script

A Python monitoring application runs from the designated programming workstation and communicates with the SDN controller through its REST API. The application authenticates with the controller, obtains an API ticket and requests the managed-device inventory. It then reports information about discovered devices, including hostname, management IP address, device type, reachability and collection status. This provides a programmable method of monitoring the SDN-managed infrastructure in addition to the controller's graphical management interface.

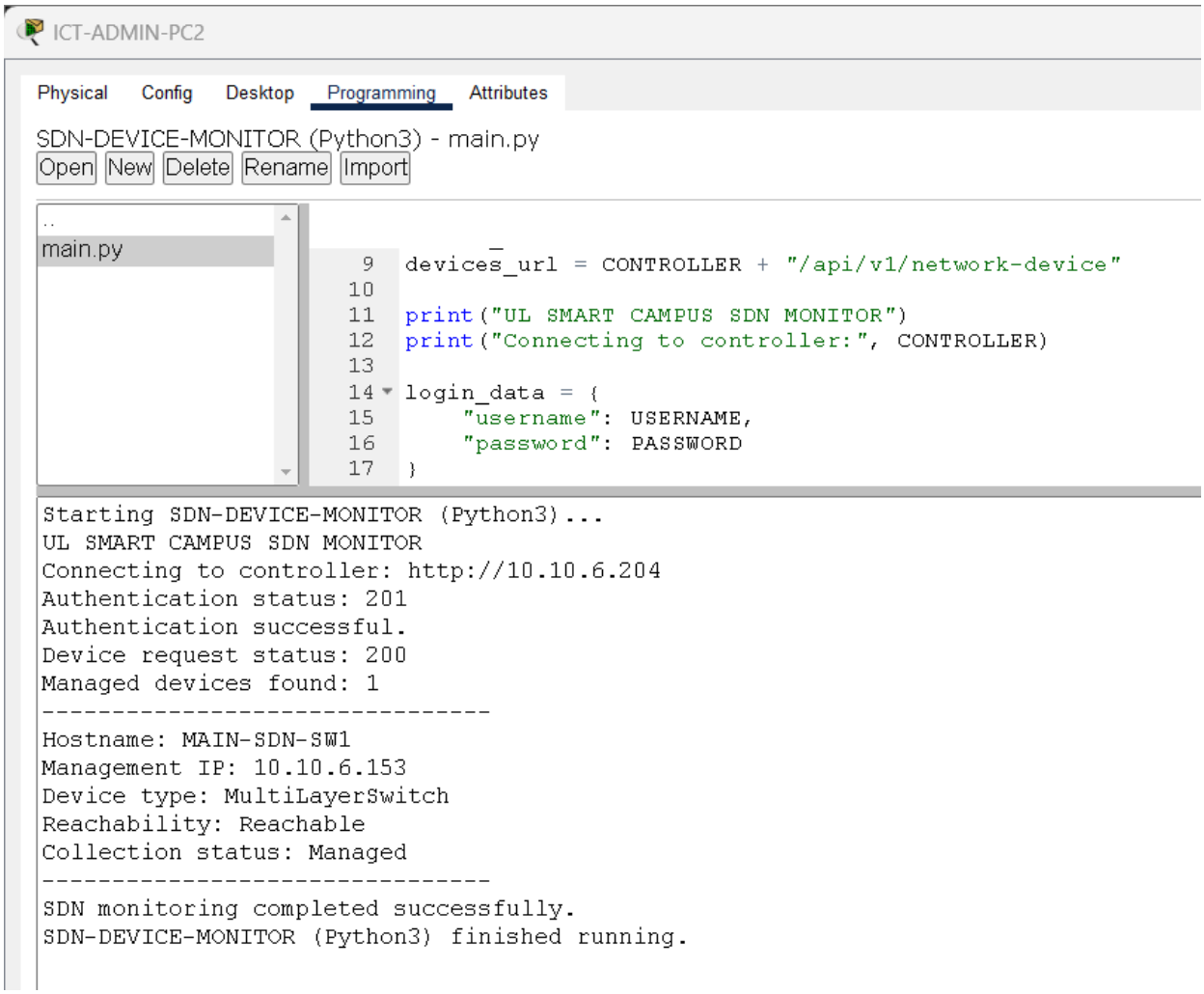


Figure 12.2: Python SDN monitoring application successfully retrieving managed-device information from the network controller API.

13 Testing Troubleshooting and Performance Analysis

13.1 Final Test Plan and Results

ID	Test	Method	Observed result	Status
T01	DHCP allocation	Renew clients in representative VLANs	Correct subnet, gateway and DNS assigned after simulator convergence	Pass
T02	Inter-VLAN routing	Ping permitted gateways and services	Permitted destinations reachable	Pass
T03	Student isolation	Attempt access to protected internal ranges	ACL denied prohibited traffic	Pass

T04	HSRP failover	Interrupt active-core path and observe states	Standby became Active; preferred core recovered with preemption	Pass
T05	STP redundancy	Inspect forwarding and alternate uplinks	Loop-free topology with alternate path retained	Pass
T06	VPN establishment	Inspect IKE and IPsec SAs and generate protected inter-campus traffic	IKE reached QM_IDLE; IPsec SAs established and packet counters increased	Pass
T07	Wireless 802.1X	Connect with valid and invalid accounts	Valid account connected; invalid account denied	Pass
T08	Residence Wi-Fi	Join residence SSID and request DHCP	Client received a 10.10.8.0/22 address	Pass
T09	VoIP registration	Inspect ephones at both sites	Phones registered with correct extensions	Pass
T10	Inter-campus call	Place call between extensions 2002 and 3001	Call established successfully	Pass
T11	QoS policy	Inspect marking and WAN policies	Voice identified/marked; strict priority policies attached	Pass with simulator limitation
T12	IoT registration	Use ICT browser to inspect registration service	Authorized IoT devices visible	Pass
T13	SSH management	Ping and SSH to managed device	Reachability confirmed; SSH enabled where configured	Pass
T14	NTP and logging	Inspect clock and server configuration	NTP and syslog services configured to use 10.10.6.203	Pass

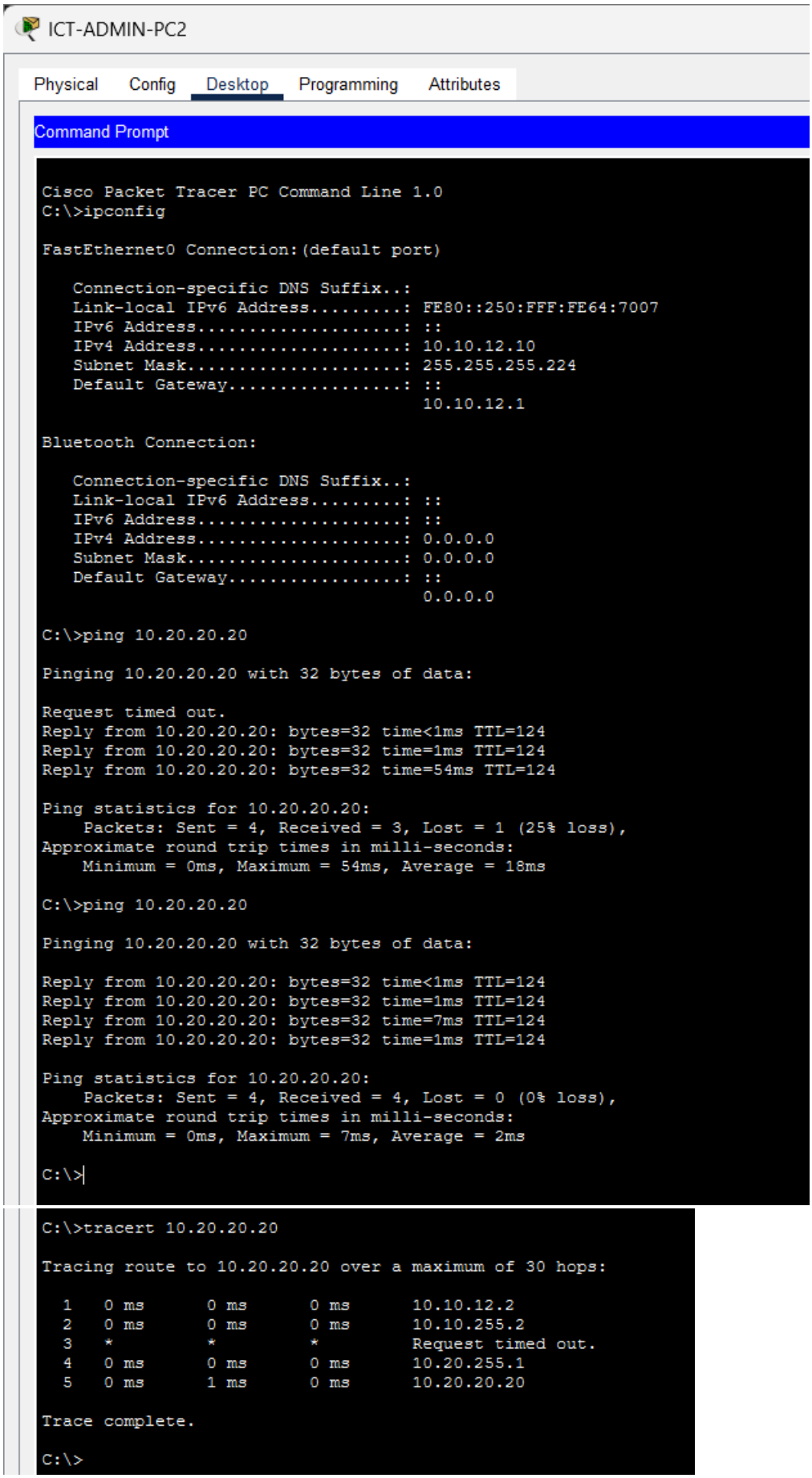


Figure 13.1(a): End-to-end connectivity and route verification from ICT-ADMIN-PC2 at the main campus to SAT-STAFF-PC1 at the satellite campus.

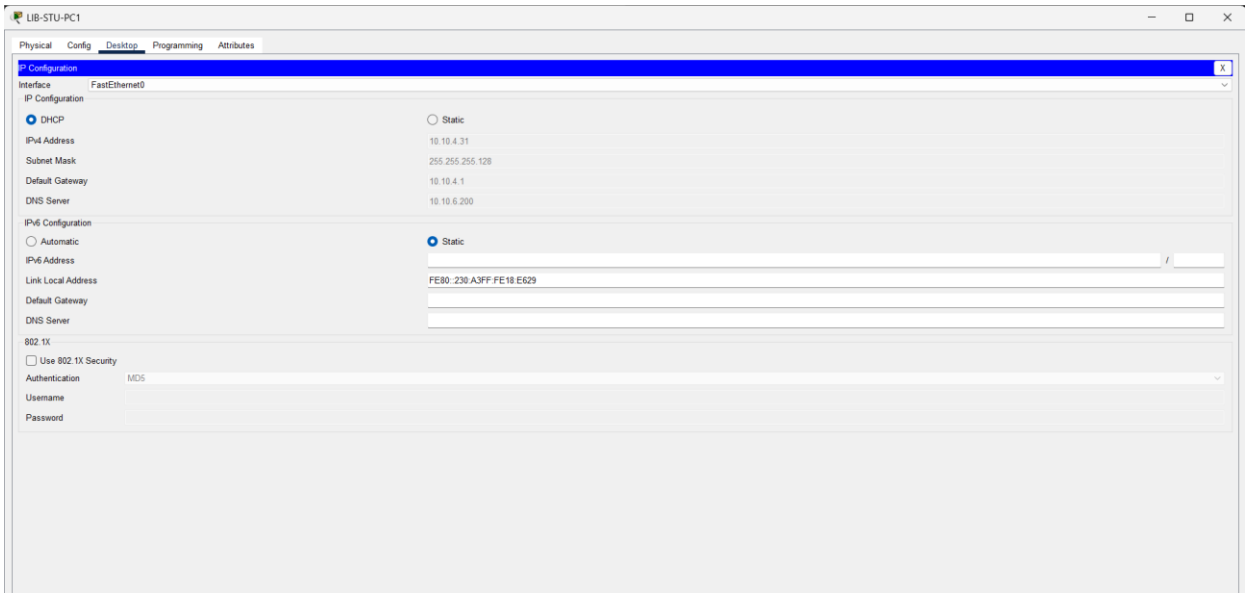


Figure 13.1(b): DHCP verification on LIB-STU-PC1 showing automatic assignment of the correct student VLAN address, subnet mask, default gateway and DNS server.

13.2 Troubleshooting Record

Problem	Root cause	Corrective action	Outcome
VLAN absent from forwarding list	VLAN not created or STP still converging	Create VLAN, permit it end-to-end and wait for forwarding	Traffic restored
Residence LAP offline	Native VLAN inconsistency or stale simulator state	Verify trunk/native VLAN, MAC learning, DHCP and restart Packet Tracer	LAP joined controller
Wireless client had no address	SSID/VLAN mapping or delayed DHCP state	Correct WLAN mapping, reselect SSID and renew DHCP	Valid address obtained
Research 802.1X failed	Endpoint 802.1X option disabled	Enable 802.1X security and retry credentials	Authentication succeeded
RADIUS ping returned unreachable	Access policy or source-path behavior	Verify management SVI, ACL permission and RADIUS settings	Authentication traffic permitted
HSRP dual-active event	Temporary loss of hello communication during topology changes	Restore VLAN path and verify STP/trunks	Normal Active/Standby state restored
WAN voice counters remained low	Packet Tracer IPsec and RTP simulation limitation	Retain DSCP EF marking before encryption and document the simulator limitation	Correct policy retained

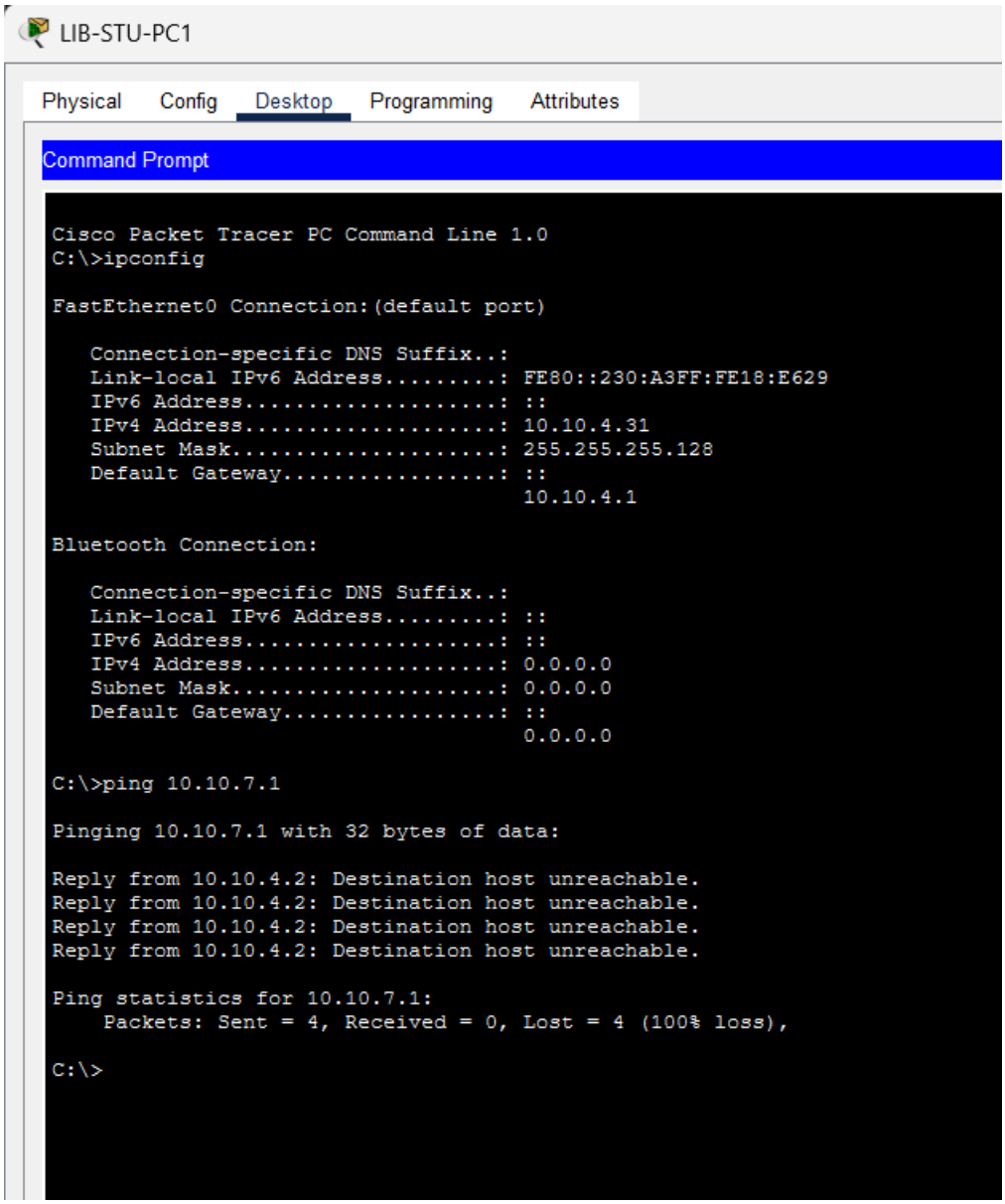


Figure 13.2: Student-network isolation test showing LIB-STU-PC1 denied access to the protected Admin VLAN.

13.3 Performance Analysis

Connectivity testing generally completed with low simulated delay. Initial packet loss observed during some first-time ping tests was attributable to address-resolution and convergence behavior, after which subsequent communication stabilized. IPsec packet counters increased in both directions during protected inter-campus communication. HSRP provided redundant default-gateway operation, while spanning tree maintained loop-free redundant Layer 2 paths. QoS policies remained attached to the WAN interfaces, although Packet Tracer's limited traffic generation and IPsec simulation restricted meaningful performance measurement of the priority queues.

The testing demonstrated that the implemented topology satisfies the project's functional requirements for segmentation, routing, redundancy, secure inter-campus communication, wireless access, voice services, IoT connectivity and centralized management within the simulation environment. However, the requirement to support more than 500 concurrent users represents a design capacity objective rather than a measured Packet Tracer load-test result. A production deployment would require wireless site surveys, measured WAN utilization, realistic concurrent traffic

generation, latency and jitter measurements, vulnerability assessment and hardware performance baselines.

14 Limitations

- Cisco Packet Tracer does not implement every Cisco IOS command, protocol feature or hardware behavior available on production equipment.
- The simulator does not fully reproduce real-world wireless propagation, interference, roaming or radio-frequency behavior.
- IPsec QoS pre-classification, DSCP handling through encrypted traffic and continuous RTP traffic generation are limited within the simulation environment.
- Performance measurements obtained in Packet Tracer do not represent physical-link utilization, realistic concurrent-user load, jitter or hardware forwarding capacity.
- The simulated SDN controller and IoT services provide a simplified subset of the capabilities available in production platforms.

15 Conclusion

The project produced a functional and secure multi-site smart campus network prototype that satisfies the main design objectives. The main and satellite campuses are interconnected through a site-to-site IPsec VPN, while users, devices and services are segmented through VLANs and ACLs. Redundant Layer 2 and Layer 3 mechanisms improve network availability, while centralized DHCP, DNS, RADIUS, monitoring, IoT and SDN services support consistent operation across the environment.

The implementation also demonstrates enterprise wireless authentication, registered lightweight access points, inter-campus VoIP communication and QoS policy enforcement. Testing and troubleshooting demonstrated the dependencies between VLAN configuration, trunking, spanning tree, DHCP, authentication, routing and application services. The final test results confirmed successful operation of the major network functions, including inter-VLAN connectivity, gateway redundancy, wireless authentication, secure inter-campus communication, voice services, IoT registration and centralized network management.

Within the documented limitations of Cisco Packet Tracer, the completed prototype demonstrates that the proposed architecture can provide a secure, segmented, redundant and centrally managed smart campus network for the University of Limpopo. The project therefore achieved its primary functional and security objectives within the scope of the simulation environment.

16 Individual Contribution Statement

This section summarizes the individual contributions of each group member to the design, implementation, testing and documentation of the smart campus network. The responsibilities listed reflect the primary tasks assigned to and completed by each member during the project.

Member name	Student number	Responsibilities and deliverables
RAEDANI NF	202305783	DHCP SETUP

RAKGETSE PI	202204262	VPN SETUP
RAMAHLO RS	202391068	QoS Research and Configuration
RAMOLOTO CB	240000760	Firewall Configuration
RAMPHADI KM	240036726	TOPOLOGY DESIGN
RANHLAKGWE MM	202379376	**Diagrams**
RAPOTU K	202173081	DNS CONFIGURATION
RASIVHETSHELE M	240040718	IoT Configuration
RAVHURA TC	202322062	Wireless 802.1X Configuration
SAMBO NL	202409799	Subnet Planning
SEKGOBELA TM	240015914	**VLAN Configuration**
SEKOTO B	202103276	VLSM Planning
SEKUBA MM	240036268	IOT CONFIGURATION
SEMENYA M	202374221	Documentation

17 References

- Cisco Systems, Cisco IOS Security Configuration Guide: Secure Connectivity and VPN.
- Cisco Systems, Catalyst Switch Configuration Guides: VLANs, STP, HSRP, Port Security and 802.1X.
- Cisco Systems, Enterprise QoS Solution Reference Network Design Guide.
- IEEE, IEEE Std 802.1X: Port-Based Network Access Control.
- National Institute of Standards and Technology, SP 800-207: Zero Trust Architecture, 2020.
- University of Limpopo, Department of Computer Science, SCOB032 Computer Networks Semester Project Brief, 2026.

Appendix A Key Verification Commands

- `show interfaces status`
- `show interfaces trunk`
- `show vlan brief`
- `show spanning-tree vlan <vlan-id>`
- `show standby brief`
- `show ip route`
- `show access-lists`
- `show crypto isakmp sa`
- `show crypto ipsec sa`
- `show policy-map interface <interface>`
- `show ephone`
- `show ip interface brief`
- `show mac address-table interface <interface>`
- `show running-config | section <section>`

- `show port-security`
- `show port-security interface <interface>`

Appendix B Evidence Register

Evidence ID	Screenshot or output	Report section
E01	Complete physical topology	3 and 4
E02	VLAN and trunk verification	5 and 6
E03	HSRP Active/Standby states	6
E04	STP root and blocked alternate path	6
E05	ACL counters and isolation test	7
E06	WLC LAP and WLAN status	8
E07	Wireless client DHCP and authentication	8
E08	IKE and IPsec security associations	9
E09	Registered ephones and connected call	10
E10	QoS marking and WAN policies	10
E11	IoT registration page	11
E12	SDN controller inventory and script output	12
E13	DHCP, permitted connectivity and traceroute results	13
E14	E14 Port security, PortFast and unused-port security	7